Supporting Information
For
Leveraging Retrieval Augmented Generation to Accelerate Discoveries on Mealworm Larvae and Plastic Degradation

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Summary:

7 pages, including 3 Figures: Knowledge graph growing (S1), Knowledge graph comparison by entity types (S2), Detailed example of ExpQ score calculation (S3)

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Supporting information S1: Knowledge graph growing

The LightRAG framework supports live insertion of new literature without retraining, underscoring RAG's inherent scalability and paving the way for specialized "growth" of the knowledge graph on enzymes, proteins, or metabolites by identifying connections between newly added data and previous knowledge base **(Figure S1)**. Such a knowledge base can be expanded to include other known degraders besides the yellow mealworms to inspire plastic biodegradation research.

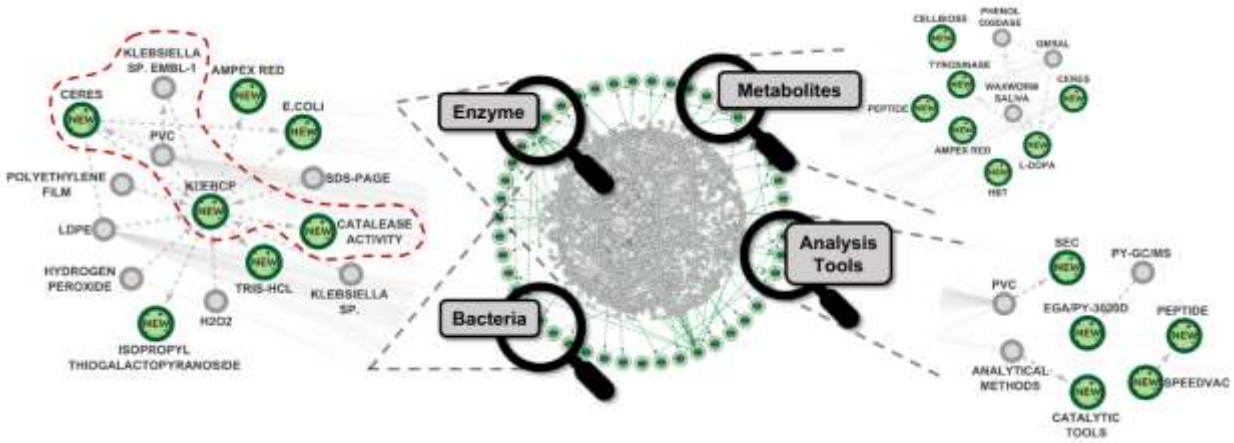


Figure S1. Illustration of the grow-ability of the knowledge graph, by adding new findings on plastic digestive enzymes. KlebCP, a catalase-peroxidase encoded by *Klebsiella* sp. EMBL-1, an isolate derived from the larval gut of *Spodoptera frugiperda*, has been shown to depolymerize PVC as a sole energy source. *Klebsiella* has been reported in gut microbiome studies for mealworms eating PVC (polyvinyl chloride).

Supporting information S2: Knowledge graph comparison by entity types

The attributes of each node, including the entity type, description and name should be present in the node table. We sub-grouped the major plastics, PS and PE, from the entire networks generated from 75 papers. The top five entity types for PS were "Bacteria", "Metabolites", "Research institution", "Plastic type", and "Experiment condition". PE had a different set of top 5 and its "Organism" had a higher frequency than "Experiment condition". Besides, the frequency of each entity type in the PS cluster was always larger than in the PE cluster **(Figure S2A)**. The abstracts collected by April 2025 were used to generate knowledge networks for

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different worm species, mealworms and greater waxworms. The entity types "Metabolites", "Experiment condition", "Research institution", "Analytical tools", and "Experiment design" were the top 5 for the mealworm. The waxworm had different top entity types and the frequencies of "Bacteria" and "Research method" were higher than "Metabolites" and "Analytical tools". Although the frequency of "Metabolites" and "Enzymes" in the mealworm cluster was larger than in the waxworm cluster, the entity types "Gene", "Pathways", and "Fungus" had lower frequencies in the former **(Figure S2B)**. When zooming into the node list of "Metabolites", we found that a few components describing that entity type, such as "frass", were also included besides the specific names of metabolites. Such limitations should be further addressed in future work, and the accurate assignment of entity types needs more careful classification and expert supervision.

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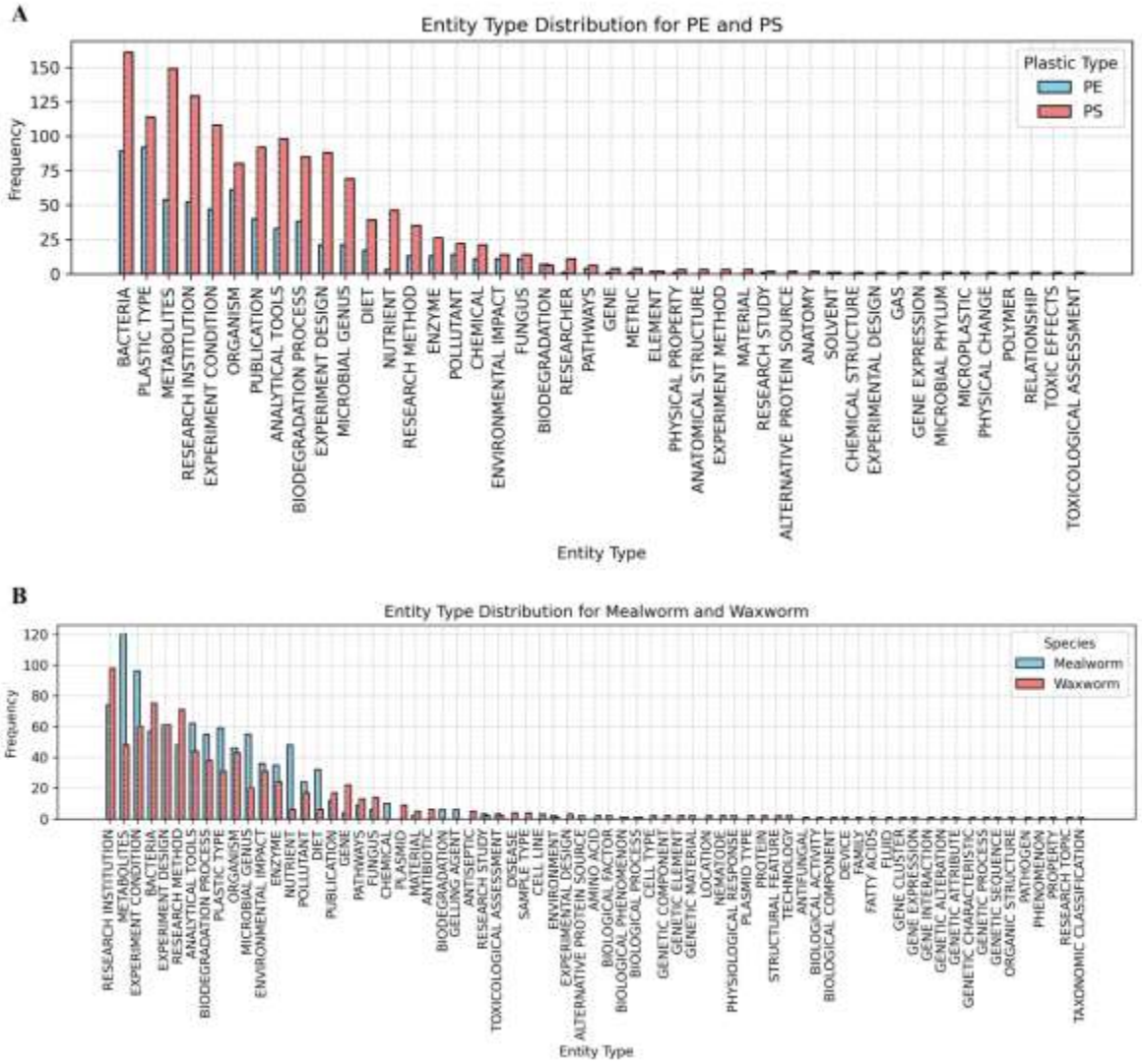


Figure S2. Comparison of entity types for **(A)** PE vs PS and **(B)** Mealworm vs Waxworm.

Our study investigated a young field with little well-degreed knowledge. Despite very few studies aimed at metabolites, enzyme activities, and gene expression, some important enzymes involved are extracted by the knowledge graph. The insect host secretes emulsifying agents and digestive enzymes that increase the plastic's bioavailability, while its symbiotic gut bacteria contribute specialized hydrolases and oxidative enzymes to cleave the polymer chains[1]. In polystyrene degradation, for example, initial oxidative attacks by enzymes (e.g. oxygenases and cytochrome P450 monooxygenases) are hypothesized to break the long polymer into smaller oligomers and styrene monomers, which are subsequently oxidized by styrene

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monooxygenase and further processed by aromatic-ring cleaving enzymes (such as laccases or peroxidases)[2]. The resulting metabolites are funneled into central metabolic pathways, notably the fatty acid β-oxidation cycle, enabling the larva to assimilate and mineralize part of the plastic as energy and biomass[3, 4]. Empirical studies captured by the knowledge graph show that mealworms can mineralize nearly half of the ingested polystyrene into CO_2, with the remainder fragmented and egested as fecal residues[3]. A few metabolic pathways of PS degradation were shown as well, but the key enzyme reactions of PE biodegradation have not yet been determined. Although a few enzymes purified from greater waxworm were claimed to be capable of degrading PE, the knowledge graph points out that assessment on the plastic degradation should be done with proper control groups and careful interpretation of FTIR spectra to avoid the false-positive results[5]. Correspondingly, *in vivo* assays have detected elevated activities of specific gut enzymes (including a C8 esterase/lipase and an esterase C4) in plastic-fed mealworms, implicating these enzymes in polymer breakdown[1]. Collectively, the knowledge graph indicates that mealworms degrade plastics via oxidative and hydrolytic enzymes that depolymerize polymers into assimilable fragments, with microbial and host enzymes cooperating to convert these fragments into CO_2 and biomass.

Supporting information S3: Detailed example of ExpQ score calculation

After coloring the key text chunks, we determined that there were five experiments (E_m in **Eq. 2**) and four data analyses (A_m in **Eq. 2**) in the experiment plan generated by LightRAG-trained GPT-5 **(Figure S3)**. Next, we counted and summed up "details" (defined in Materials and Method section 2.5): there were sixteen details for Experiment Design section (D_m^E in **Eq. 2**) and fifteen details for Data Analysis section (D_m^A in **Eq. 2**). For each model (m), those counts as numerators in **Eq. 2** were recorded. Then, the maximum across the entire set of models (M)

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were determined as denominators in **Eq. 2**. Finally, ExpQ score for LightRAG-trained GPT-5 was calculated according to **Eq. 2**.

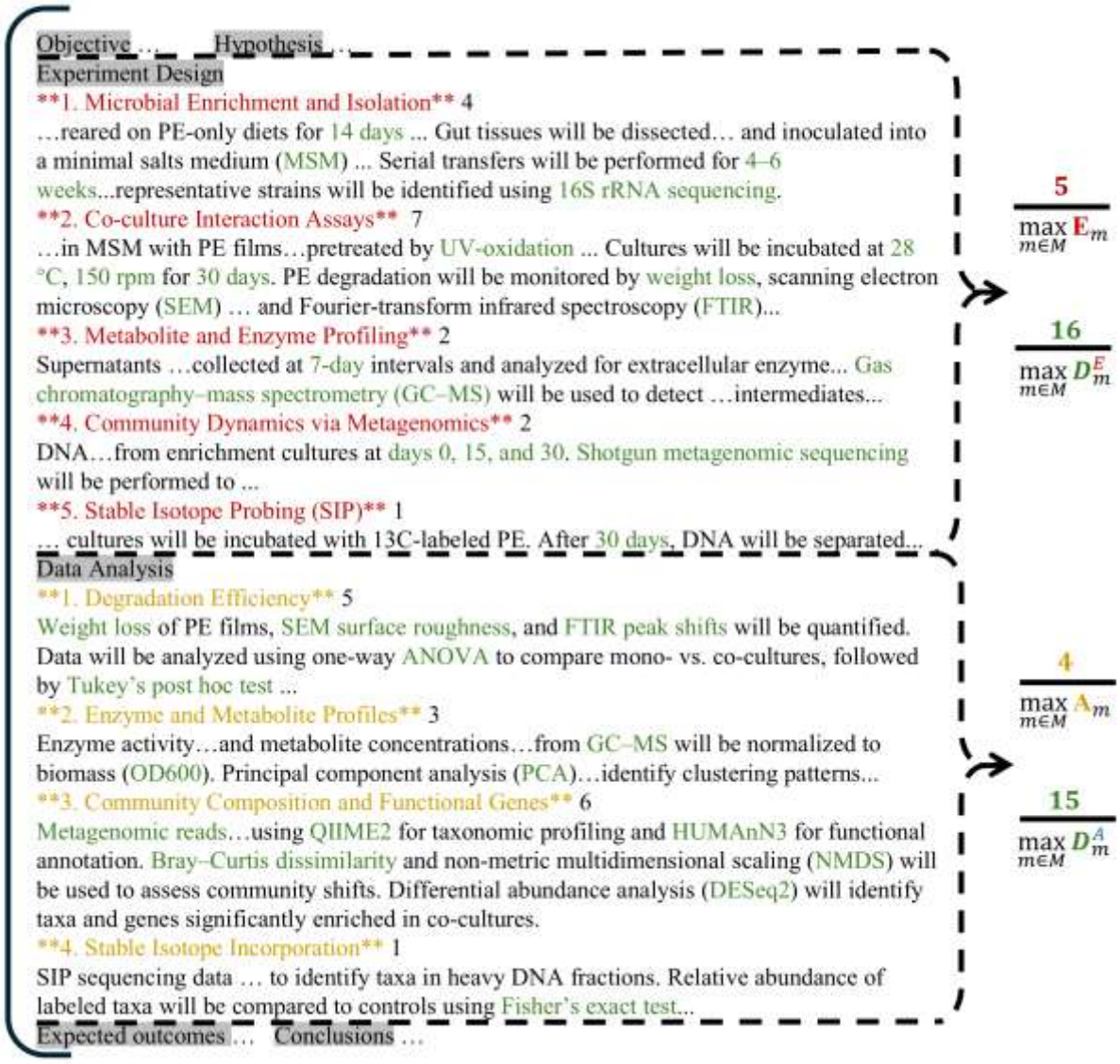


Figure S3. Explicit process of determining the numerators in Eq. 2 for ExpQ score calculation. In the briefed model output of LightRAG-trained GPT-5, experiments were colored red and data analyses were colored yellow. E_m in **Eq. 2** as the number of experiments was determined for this model. Similarly, A_m in **Eq. 2** as the number of data analyses was also determined. All the "details" colored as green and counted. The number of "details" for each experiment was noted and then summed up as D_m^E in **Eq. 2** for the Experiment Design section. Similar steps were applied for determining D_m^A in **Eq. 2**.

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